Fiscal Year:	FY 2011	**Task Last Updated:**	FY 10/27/2010
PI Name:	Bailey, Susan M. Ph.D.		
Project Title:	Telomeric proteins in the radiation/DNA damage response		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Fort Collins	**State:**	CO
Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date is now 12/31/2012 per NSSC information (Ed., 12/21/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Liber, Howard (Colorado State University)		
Grant/Contract No.:	NNX08AB65G		
Performance Goal No.:			
Performance Goal Text:			
	This research addresses Radiation Health Bioastronautics Roadmap (BR) Carcinogenesis Questions 28c, d and j, and is aimed at advancing current knowledge and mechanistic understanding of the DNA damage processing that underlies radiation effects in general, and space radiation in particular. The goal of the studies proposed here is to provide information that will aid identification of susceptible individuals, ultimately improving risk prediction for radiation carcinogenesis in astronauts exposed to high energy charged particles in space. Summary: Appropriate cellular response to natural chromosomal termini (telomeres) and to exogenously created double-strand breaks (DSBs) is critical to maintaining genomic stability. Incorrect handling of DSB ends leads to mutation, chromosomal rearrangements, instability and carcinogenesis. Altered levels of telomeric proteins have been reported in some tumors. The interplay between telomeres and DSBs may be better understood not only by viewing the striking differences between them, but		

Task Description:	perhaps even more importantly by viewing the obvious similarity-they are both DNA ends, the very substrate both telomere and damage response/repair proteins specifically recognize and bind. We have been investigating the role DNA repair proteins play at telomeric ends. Recent and intriguing studies suggesting that the classic telomere-binding protein TRF2 is also found (rapidly and transiently) at sites of photo-induced DNA DSBs, has turned the question around-what are telomere proteins doing at DSB ends? Here, we propose to address this fascinating question by investigating the role(s) telomeric proteins play in ionizing radiation (IR)-induced DNA damage response/processing/repair. Our approach assesses mutation frequency and spectra at the molecular level, as well as extends the analyses to the cytogenetic level. Our preliminary studies lend support to the view that telomeric proteins aren't just for telomeres anymore. Increasing our understanding of previously unrecognized relationships between the radiation/DNA damage response and telomere function will facilitate predicting the carcinogenic risk of exposure to radiation in general, and to astronauts exposed to HZE particles in particular. Ultimately, we foresee utilizing telomere instability/dysfunction as a useful biomarker of underlying repair deficiency in order to identify individual susceptibility to radiation carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our investigation of interdependencies between DNA repair and telomeres will ultimately lead to identification and characterization of previously unappreciated telomere dysfunction, which in turn will provide a useful indication of an underlying repair defect. Therefore, telomere instability may ultimately serve as a sensitive biomarker, identifying individual susceptibility to radiation carcinogenesis in general and individual susceptibility to HZE exposure in particular.
Task Progress:	Appropriate cellular response to natural chromosomal termini (telomeres) and to exogenously created double-strand breaks (DSBs) is critical to maintaining genomic stability. Incorrect handling of DSB ends leads to mutation, chromosomal rearrangements, instability and carcinogenesis. Altered levels of telomeric proteins have been reported in some tumors. The interplay between telomeres and DSBs may be better understood not only by viewing the striking differences between them, but perhaps even more importantly by viewing the obvious similarity – they are both DNA ends, the very substrate both telomere and damage response/repair proteins specifically recognize and bind. We have been investigating the role DNA repair proteins play at telomeric ends [1]. Recent and intriguing studies suggesting that the “classic” telomere-binding protein TRF2 is also found (rapidly and transiently) at sites of photo-induced DNA DSBs, has turned the question around – what are telomere proteins doing at DSB ends? We have recently shown that ionizing radiation (IR)-induced DSBs are insufficient to initiate TRF2 recruitment to damage sites [2]. We are now working to further address this fascinating question by investigating the role(s) telomeric proteins play in IR-induced DNA damage response/processing/repair. Our overall aim is to study the effects of depleting expression of key telomeric binding proteins on DNA damage response/repair. The telomeric DNA-binding proteins TRF1 and TRF2 have multiple interacting binding partners, including TIN2, POT1 and tankyrase1. Many interactions between these players (and others) have been reported. Interestingly, TIN2 has recently been shown to have novel extra-telomeric functions in nuclear organization. Telomerase (hTERT) is also known to interact with many telomeric as well as non-telomeric proteins (including DNA repair proteins), and extra-curricular/pro-tumorigenic activities of telomerase independent of its classical role in telomere length maintenance - including enhanced repair - have been proposed, but remains an open issue. Our approach is to knockdown expression of these critical telomeric proteins in human cell lines utilizing siRNA strategies. DNA damage is induced by exposure to ionizing radiations (low LET gamma (gamma-rays or HZE 56Fe ions). We then employ a powerful combination of mutational analyses and cytogenetic approaches to examine the phenotypic and genotypic consequences of the telomeric deficiencies on DNA repair endpoints. Initial indications are that small interfering RNA (siRNA) knockdown of TRF2 results in a significant increase in radiation-induced mutation at the heterozygous thymidine kinase (tk) locus in human lymphoblastoid cells. Depletion of the Protection of Telomeres (POT1) protein also resulted in increased radiation mutagenesis. Both TRF2 and POT1 knockdowns behaved similarly, qualitatively and quantitatively; each led to elevated background mutant fractions, and also to enhanced radiation mutagenesis. Additional evidence that POT1 plays a critical role in DNA damage response/repair and tumorigenesis was presented in our manuscript demonstrating that mouse Pot1 deficiency promotes telomeric recombination, chromosomal instability and malignant transformation in vivo [3]. Intrigued by the dynamics of the seemingly contradictory yet integrated cellular responses to the requisites of preserving telomere integrity while also efficiently repairing damaged DNA, we next investigated roles of the telomere associated Poly ADP ribose polymerase (PARP), tankyrase 1, in both telomere function and the DNA damage response following exposure to ionizing radiation. Tankyrase 1 siRNA knockdown in human cells significantly elevated recombination specifically within telomeres; i.e., telomere sister chromatid exchange (T-SCE), a phenotype with the potential of accelerating cellular senescence, and therefore aging [4]. Most surprising however, was that depletion of tankyrase 1 resulted in concomitant and rapid reduction of the non-homologous end-joining protein DNA-PKcs, while Ku86 and ATM protein levels remained unchanged; DNA-PKcs mRNA levels were also unaffected. We found that the requirement of tankyrase 1 for DNA-PKcs protein stability reflected the necessity of its PARP enzymatic activity. We also demonstrated that depletion of tankyrase 1 resulted in proteasome-mediated DNA-PKcs degradation, explaining the associated defective damage response observed; i.e., increased sensitivity to ionizing radiation-induced cell killing, mutagenesis, chromosome aberration and telomere fusion. We provided the first evidence for regulation of DNA-PKcs by tankyrase 1 PARP activity and taken together, identified roles of tankyrase 1 with implications not only for DNA repair and telomere biology, but also for cancer and aging. The following articles have resulted from this work. Regulatory roles of tankyrase 1 at telomeres and in DNA repair: suppression of T-SCE and stabilization of DNA-PKcs. Ryan C. Dregalla, Junqing Zhou, Rupa R. Idate, Christine L.R. Battaglia, Howard L. Liber and Susan M. Bailey. Currently in press, AGING. An invited News and Views article also highlights these findings, and puts them in perspective with our recent WRN/BLM/ERCC1 PNAS paper: The telomere protein tankyrase 1 regulates DNA damage responses at telomeres. Sandy Chang, Dept. of Laboratory Medicine, Yale University School of Medicine. Currently in press, AGING. We were also invited to contribute a Research Perspective regarding our WRN/BLM/ERCC1 PNAS paper: Telomere Sister Chromatid Exchange and the Process of Aging Krastan B. Blagoev, Edwin H. Goodwin and Susan M. Bailey.

	AGING, Sep, 2010. References [1] Williams ES, Klingler R, Ponnaiya B, Hardt T, Schrock E, Lees-Miller SP, Meek K, Ullrich RL, Bailey SM. Telomere dysfunction and DNA-PKcs deficiency: characterization and consequence. Cancer Res 2009;69 (5):2100-7. [2] Williams ES, Stap J, Essers J, Ponnaiya B, Luijsterburg MS, Krawczyk PM, Ullrich RL, Aten JA, Bailey SM. DNA double-strand breaks are not sufficient to initiate recruitment of TRF2. Nat Genet 2007;39 (6):696-8; author reply 8-9. [3] Wu L, Multani AS, He H, Cosme-Blanco W, Deng Y, Deng JM, Bachilo O, Pathak S, Tahara H, Bailey SM, Deng Y, Behringer RR, Chang S. Pot1 deficiency initiates DNA damage checkpoint activation and aberrant homologous recombination at telomeres. Cell 2006;126 (1):49-62. [4] Hagelstrom RT, Blagoev KB, Niedernhofer LJ, Goodwin EH, Bailey SM. Hyper telomere recombination accelerates replicative senescence and may promote premature aging. Proc Natl Acad Sci U S A 2010;107 (36):15768-73.
Bibliography Type:	Description: (Last Updated: 12/10/2025)
Articles in Peer-reviewed Journals	Dregalla RC, Zhou J, Idate RR, Battaglia CLR, Liber HL, Bailey SM. "Regulatory roles of tankyrase 1 at telomeres and in DNA repair: suppression of T-SCE and stabilization of DNA-PKcs." Aging (Albany NY). 2010 Oct;2(10):691-708. PubMed PMID: 21037379 ; http://www.impactaging.com/papers/v2/n10/abs/100210a.html , Oct-2010
Articles in Peer-reviewed Journals	Blagoev KB, Goodwin EH, Bailey SM. "Telomere sister chromatid exchange and the process of aging." Aging (Albany NY). 2010 Oct;2(10):727-30. PMID: 20952810 , Oct-2010
Articles in Peer-reviewed Journals	Hagelstrom RT, Blagoev KB, Niedernhofer LJ, Goodwin EH, Bailey SM. "Hyper telomere recombination accelerates replicative senescence and may promote premature aging." Proc Natl Acad Sci U S A. 2010 Sep 7;107(36):15768-73. PMID: 20798040 , Sep-2010